HEART DISEASE ANALYSIS USING TABLEAU

Project Documentation Report

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I would also like to thank the **mentors and instructors of SkillWallet** for their valuable guidance and support throughout the completion of this project.

I would like to appreciate the efforts and cooperation of my **team members**, whose support and collaboration helped in successfully completing this project. Working together as a team helped us share ideas, solve problems, and learn new analytical skills.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of this project.

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1. Introduction

Heart disease is one of the leading causes of death worldwide. Early identification of risk factors and patterns related to heart disease can help in better prevention and treatment.

With the help of data analytics and visualization tools such as Tableau, large medical datasets can be analyzed effectively. Visualization helps transform raw data into meaningful insights that can be easily understood.

In this project, the heart disease dataset is analyzed to identify patterns related to various health parameters such as age, cholesterol level, blood pressure, and gender.

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2. Problem Statement

Medical datasets contain large amounts of information that can be difficult to interpret in raw form. The main problem addressed in this project is analyzing heart disease data and identifying the key factors that influence the occurrence of heart disease.

The project uses Tableau to convert raw medical data into visual dashboards and charts so that patterns and trends can be easily identified.

3. Objectives of the Project

The main objectives of this project are:

- To analyze the heart disease dataset.
- To perform data cleaning and preparation.
- To create meaningful data visualizations using Tableau.
- To design an interactive dashboard.
- To identify patterns and trends in heart disease data.
- To present insights using Tableau Story.

4. Data Collection

The dataset used in this project contains medical information about patients that helps determine whether a patient has heart disease or not.

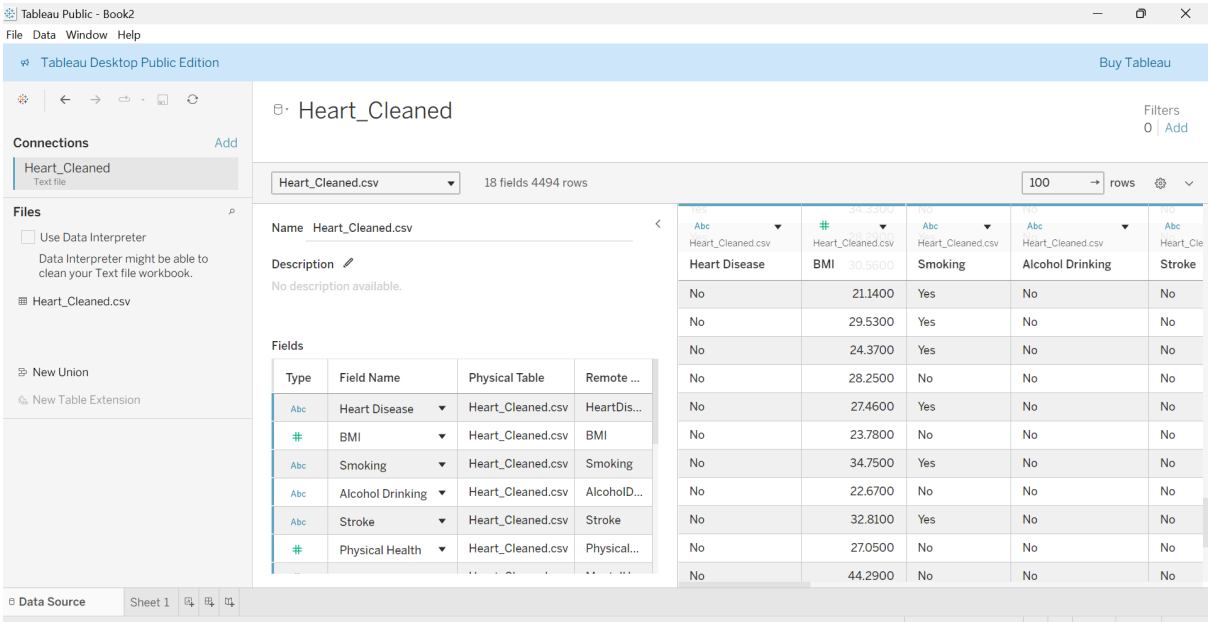
Important attributes in the dataset include:

- Age
- Sex
- Chest Pain Type
- Resting Blood Pressure
- Cholesterol Level

- Maximum Heart Rate Achieved
- Exercise Induced Angina
- Oldpeak
- Target Variable (Heart Disease Presence)

This dataset provides valuable information for analyzing risk factors associated with heart disease.

5. Data Extraction



The dataset was imported into Tableau for analysis.

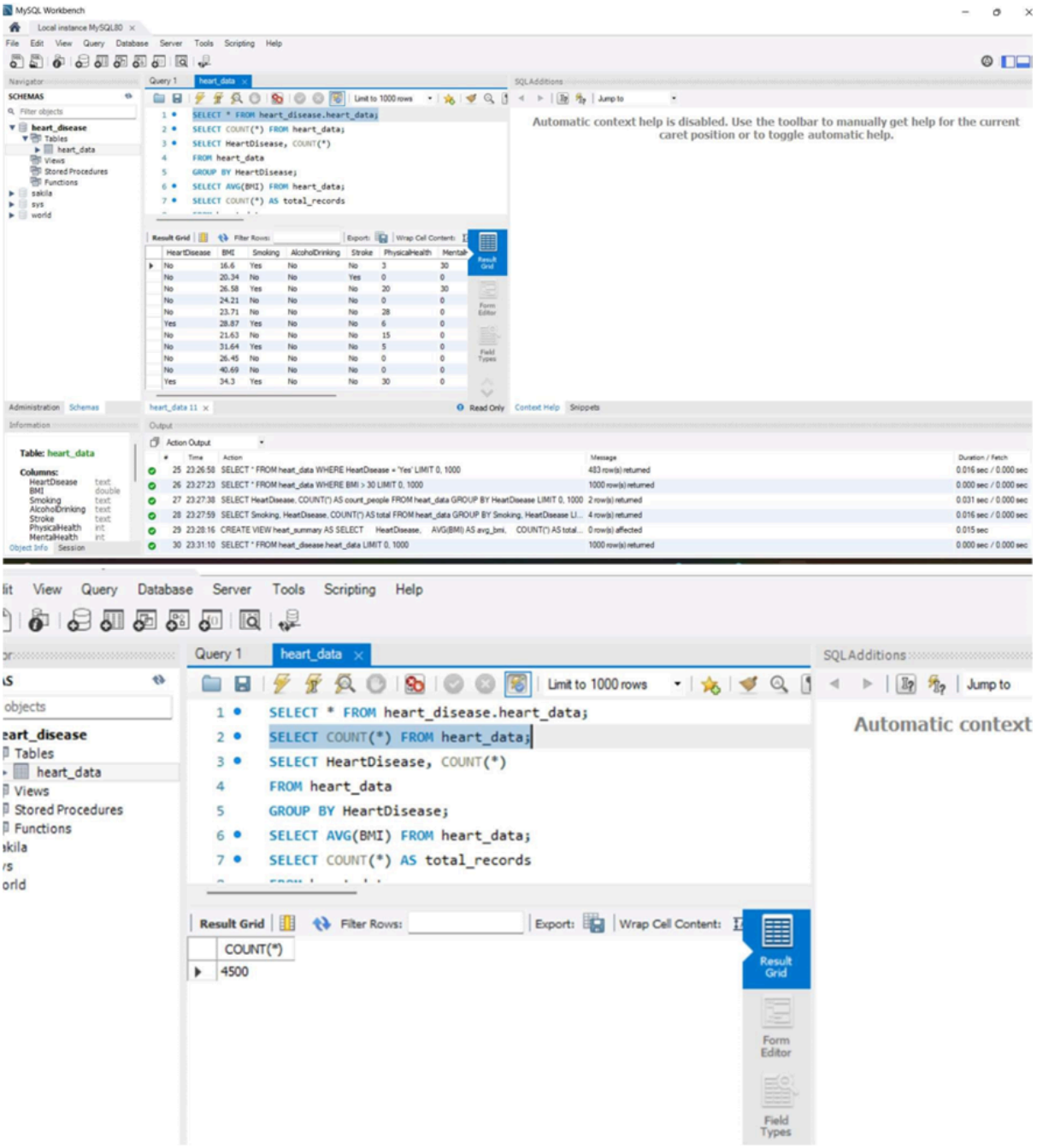
Steps involved in data extraction include:

- Importing the dataset from a CSV file or database.
- Connecting the dataset to Tableau.

- Verifying column names and data types.
- Checking that all records are correctly loaded.

This ensures that the dataset is ready for further processing.

6. Data Cleaning and Preparation

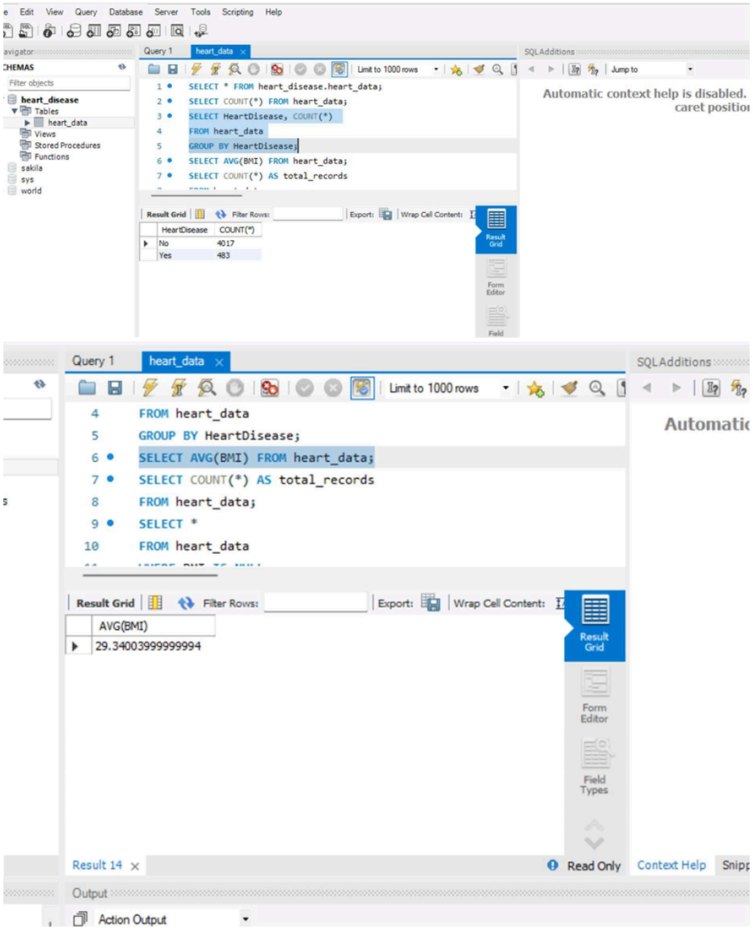


Data cleaning is an important step before performing analysis. It ensures the accuracy and reliability of the dataset.

Data preparation steps include:

- Checking and handling missing values.
- Removing duplicate records.
- Correcting incorrect data types.
- Renaming columns for better readability.
- Creating calculated fields where required.

Proper data preparation improves the quality of analysis.



7. Data Visualization

Various types of visualizations were created using Tableau to analyze the dataset.

The visualizations include:

Bar Chart: Used to analyze the relationship between age groups and heart disease occurrence.

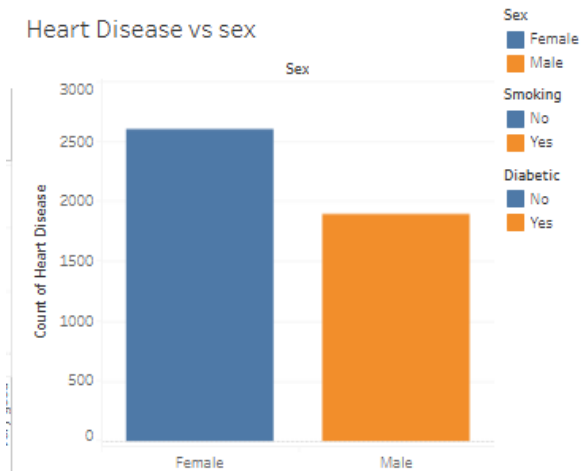
Pie Chart: Shows the distribution of male and female patients in the dataset.

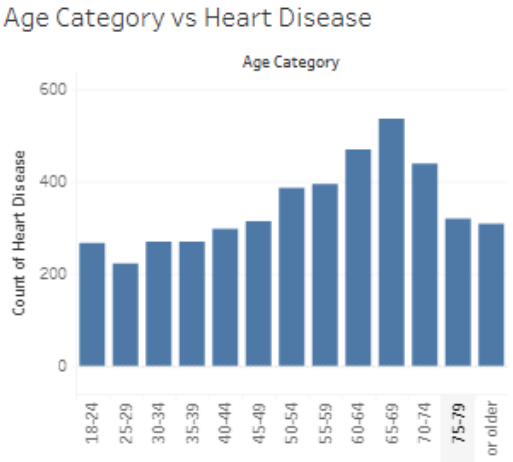
Histogram: Displays the age distribution of patients.

Line Chart: Used to observe trends in cholesterol levels.

Heat Map: Helps identify the relationship between chest pain type and heart disease.

These visualizations make it easier to understand patterns and trends in the data.





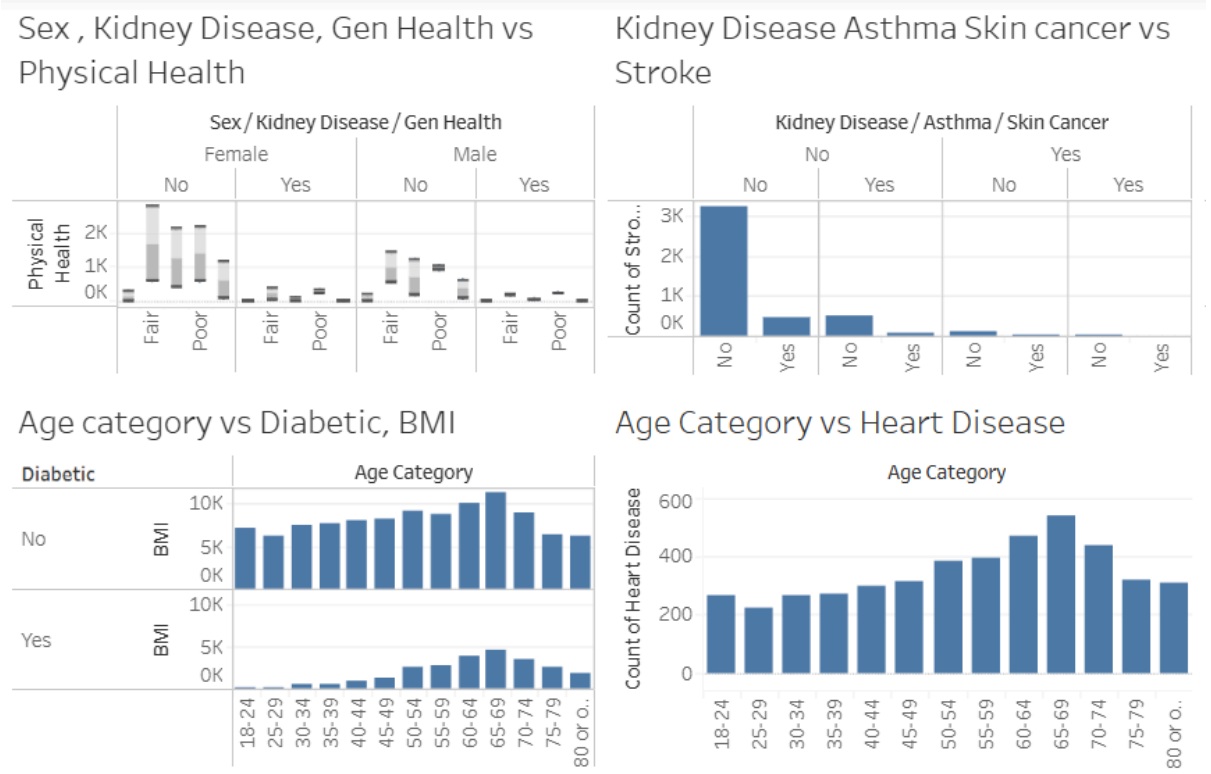
8. Dashboard Creation

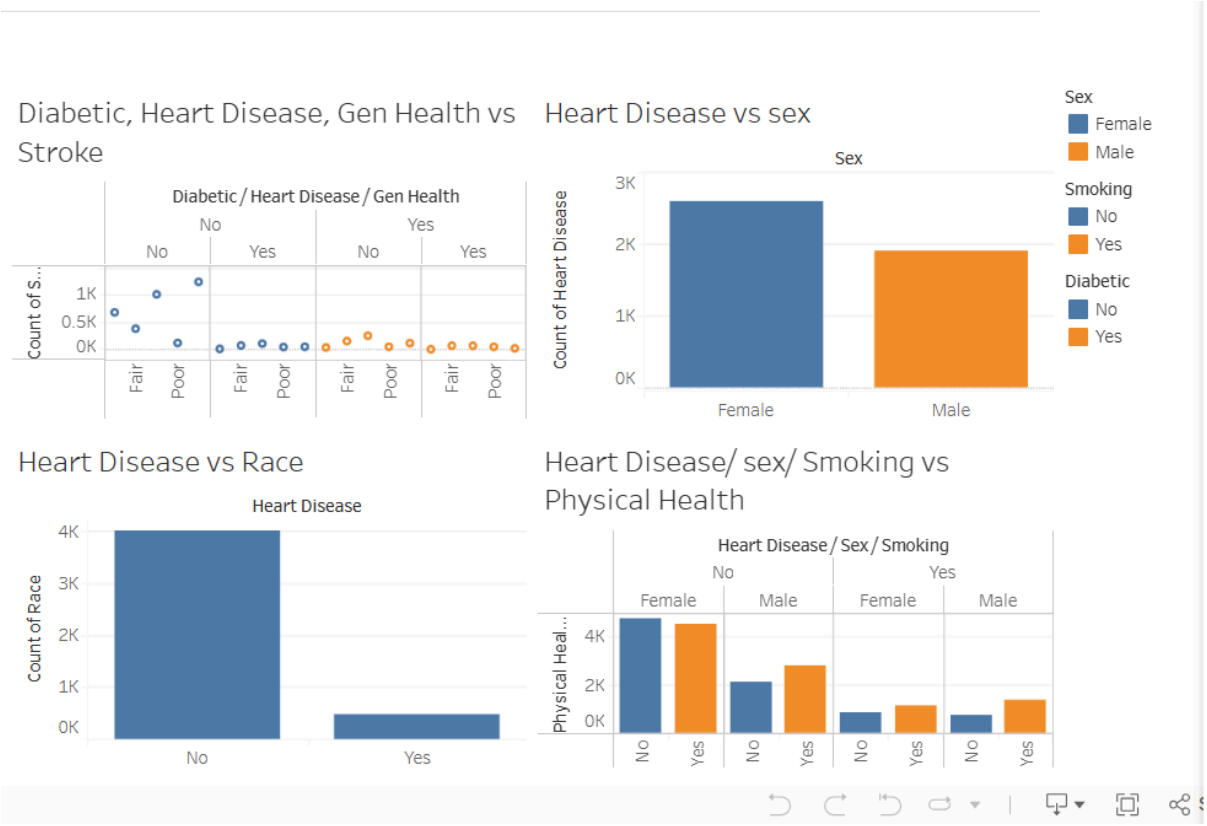
An interactive dashboard was created by combining multiple charts and visualizations.

Features of the dashboard include:

- Multiple charts displayed in a single view
- Interactive filters for better data exploration
- Clear representation of important insights
- Easy comparison of different variables

The dashboard allows users to explore the dataset and understand heart disease risk factors effectively.





9. Story Creation

Tableau Story feature was used to present the analysis step by step.

The story includes:

1. Dataset overview
2. Age-based analysis
3. Gender-based analysis
4. Cholesterol level analysis
5. Final insights and observations

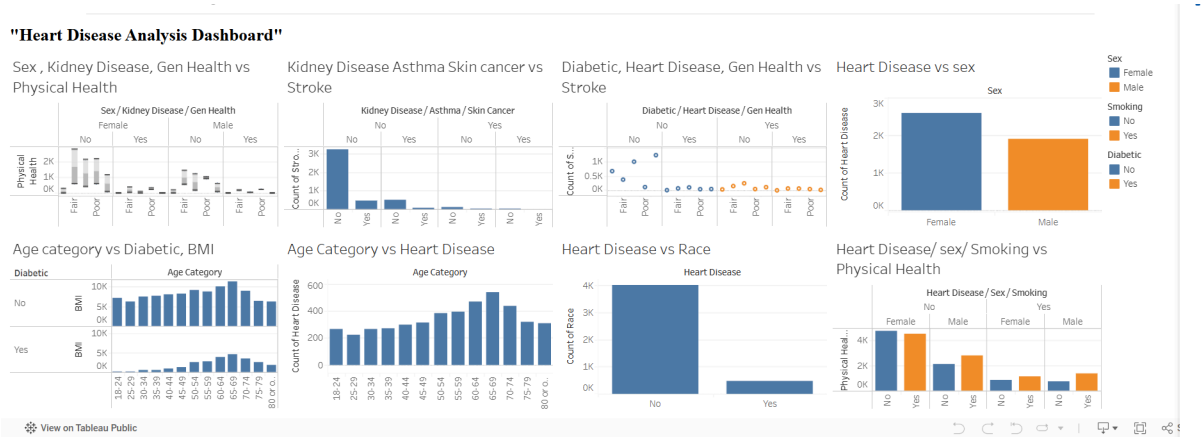
This structured approach helps in explaining the data analysis process clearly.

10. Results and Insights

After analyzing the dataset, the following insights were observed:

- Middle-aged individuals have a higher risk of heart disease.
- High cholesterol levels increase the probability of heart disease.
- Male patients have a slightly higher occurrence of heart disease compared to female patients.
- Certain types of chest pain are strongly associated with heart disease.

These insights highlight important health factors related to heart disease.



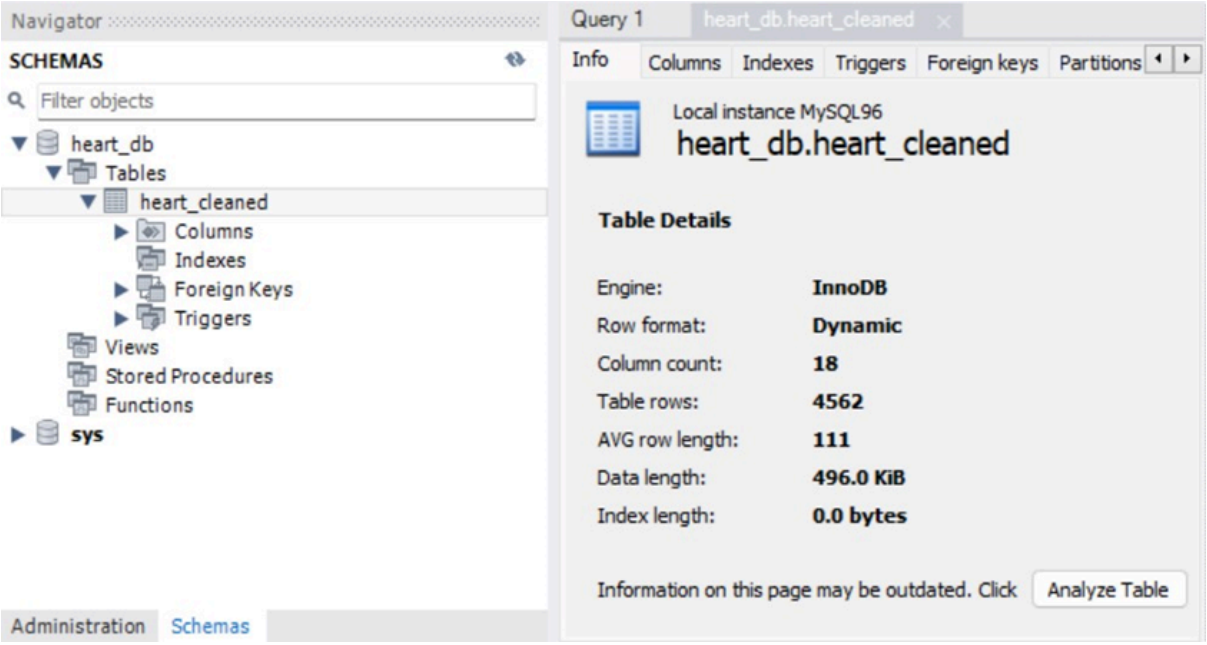
11. Performance Testing

The dashboard and visualizations were tested to ensure proper functionality.

Testing steps included:

- Verifying data accuracy
- Checking dashboard loading performance
- Testing interactive filters
- Ensuring proper visualization display

This ensures that the dashboard works efficiently and provides reliable insights.



SQL File 1*

Limit to 10

```
USE heart_db;

SELECT COUNT(*) AS total_records
FROM heart_cleaned;
```

Result Grid | Filter Rows:

total_records
4494

Result Grid

SQL File 1*

Limit to 1000 rows

```
SHOW COLUMNS FROM heart_cleaned;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
HeartDisease	text	YES		NULL	
BMI	double	YES		NULL	
Smoking	text	YES		NULL	
AlcoholDrinking	text	YES		NULL	
Stroke	text	YES		NULL	
PhysicalHealth	int	YES		NULL	

Result Grid

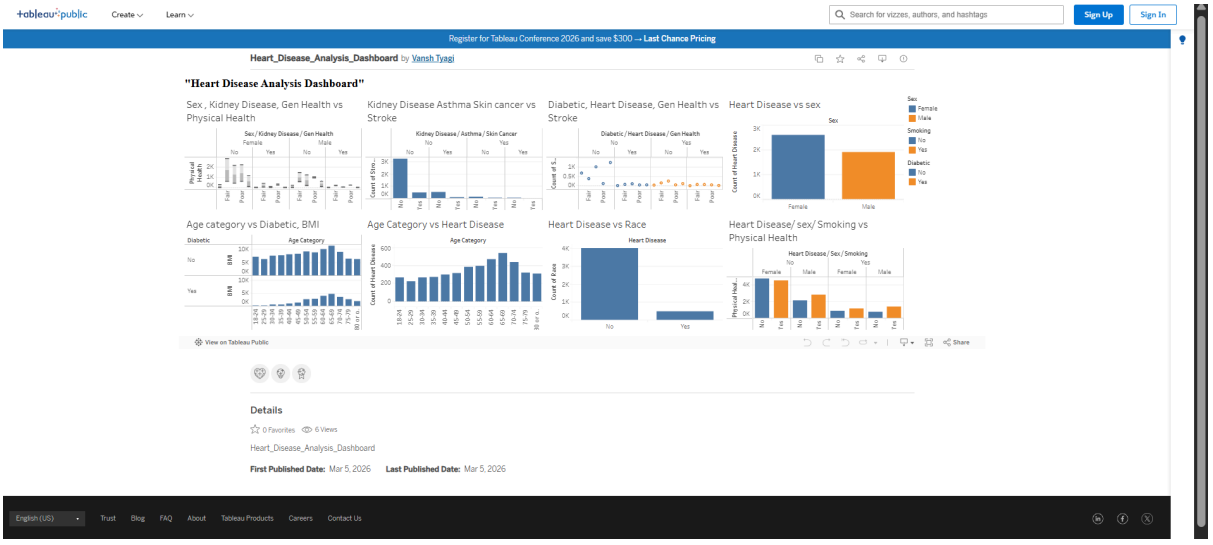
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
MentalHealth	int	YES		NULL	
DiffWalking	text	YES		NULL	
Sex	text	YES		NULL	
AgeCategory	text	YES		NULL	
Race	text	YES		NULL	
Diabetic	text	YES		NULL	

12. Web Integration

The final dashboard was published using **Tableau Public** or integrated into a web platform. This allows users to access the dashboard online and interact with the visualizations easily.

Web integration improves accessibility and makes the analysis available to a wider audience.



13. Key Findings

The following key findings were observed from the analysis:

- Heart disease risk increases with age.
- High cholesterol levels contribute to a higher risk of heart disease.
- Male patients are slightly more affected than female patients.
- Chest pain type is an important indicator for heart disease detection.

These findings can help in better understanding the major risk factors.

14. Conclusion

This project successfully analyzed the heart disease dataset using Tableau. Through data cleaning, visualization, and dashboard creation, meaningful insights were generated from the dataset.

The use of dashboards and stories helped in presenting the data in an interactive and understandable way. The analysis provides valuable information about the risk factors associated with heart disease and demonstrates the importance of data analytics in healthcare.

15. References

- Heart Disease Dataset (Public Dataset)
- Tableau Official Documentation
- SkillWallet Learning Resources